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Display Apparatus

Abstract

A display apparatus includes a display panel including a substrate in which a display area in which a plurality of pixels is disposed and a non-display area in an outer periphery of the display area are defined. The display apparatus further includes a first plate disposed below the display panel. The display apparatus further includes a black layer disposed on a bottom surface of the first plate. The display apparatus further includes a printed circuit board disposed below the first plate. The display apparatus further includes a first adhesive layer which bonds the first plate and the printed circuit board. A bottom surface of the first plate includes a center area and an edge area which encloses the center area and includes a first area and a second area and the black layer is disposed in the first area among the first area and the second area.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority of Republic of Korea Patent Application No. 10-2024-0029208 filed on Feb. 28, 2024, which is hereby incorporated by reference in its entirety.

BACKGROUND

Field

[0002] The present disclosure relates to a display apparatus.

Description of the Related Art

[0003] As it enters the information era, a field of a display apparatus which visually expresses electrical information signals has been rapidly developed and studies are continued to improve performances of various display apparatuses, such as a thin-thickness, a light weight, and low power consumption.

[0004] As exemplary display apparatus, there are a liquid crystal display (LCD), an electro-wetting display (EWD), and an organic light emitting display (OLED).

[0005] Among them, the electroluminescence display apparatus is a self-emitting display apparatus so that a separate light source is not necessary, which is different from the liquid crystal display apparatus. Therefore, the electroluminescence display apparatus may be manufactured to have a light weight and a thin-thickness. Further, since the electroluminescent display apparatus is advantageous not only in terms of power consumption due to the low voltage driving, but also in terms of color implementation, a response speed, a viewing angle, a contrast ratio (CR), it is expected to be utilized in various fields.

SUMMARY

[0006] An object to be achieved by an exemplary embodiment of the present disclosure is to provide a display apparatus which suppresses occurrence of stain defects of a display panel due to exposure to external light on a rear surface of a display panel.

[0007] Another object to be achieved by an exemplary embodiment of the present disclosure is to provide a display apparatus which guides the bending of the display panel.

[0008] Further, still another object to be achieved by an exemplary embodiment of the present disclosure is to provide a display apparatus which guides placement positions of components which are disposed on a rear surface of the display panel.

[0009] Objects of the present disclosure are not limited to the above-mentioned objects, and other objects, which are not mentioned above, can be clearly understood by those skilled in the art from the following descriptions.

[0010] In order to achieve the objects as described above, according to an aspect of the present disclosure, a display apparatus includes a display panel including a substrate in which a display area in which a plurality of pixels is disposed and a non-display area in which an outer periphery of the display area are defined; a first plate disposed below the display panel; a black layer disposed on a bottom surface of the first plate; a printed circuit board disposed below the first plate; and a first adhesive layer which bonds the first plate and the printed circuit board. At this time, a bottom surface of the first plate includes a center area and an edge area which encloses the center area and includes a first area and a second area and the black layer is disposed in the first area among the first area and the second area.

[0011] Other detailed matters of the exemplary embodiments are included in the detailed description and the drawings.

[0012] According to the exemplary embodiment of the present disclosure, in the display apparatus, a layer which suppresses exposure to external light is disposed on a rear surface of the display panel to reduce a damage of the display panel and suppress stain defects.

[0013] According to the exemplary embodiment of the present disclosure, in the display apparatus, the exposure to external light from a rear surface of the display panel is suppressed, but a panel alignment mark required to bend the display panel is exposed to guide the bending of the display panel. By doing this, misalignment of components which are disposed in a bending area is suppressed.
[0014] According to the exemplary embodiment of the present disclosure, in the display apparatus, the exposure to external light from a rear surface of the display panel is suppressed, but placement of a printed circuit board which is disposed on the rear surface of the display panel is guided to suppress misalignment.
[0015] According to the exemplary embodiment of the present disclosure, in the display apparatus, a process of attaching a separate blocking film on a rear surface of the display panel is omitted, thereby simplifying the process and reducing a production cost.
[0016] The effects of the present disclosure are not limited to the aforementioned effects, and other effects, which are not mentioned above, will be apparently understood to a person having ordinary skill in the art from the following description.
[0017] The effects of the present disclosure are not limited to the aforementioned effects, and other effects, which are not mentioned above, will be apparently understood to a person having ordinary skill in the art from the following description.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0018] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
[0019] FIG. **1** is a block diagram of a display apparatus according to an exemplary embodiment of the present disclosure;
[0020] FIG. **2** is a front view of a display apparatus according to an exemplary embodiment of the present disclosure;
[0021] FIG. **3** is a plan view of a substrate of a display panel as seen from a front side according to an exemplary embodiment of the present disclosure;
[0022] FIG. **4** is a cross-sectional view taken along the line I-I′ of FIG. **2** according to an exemplary embodiment of the present disclosure;
[0023] FIG. **5** is a plan view of a first plate as seen from a rear surface of a display panel according to an exemplary embodiment of the present disclosure;
[0024] FIG. **6**A is a plan view illustrating an example of an adhesive layer disposed below a first plate according to an exemplary embodiment of the present disclosure;
[0025] FIG. **6**B is a plan view illustrating another example of an adhesive layer disposed below a first plate according to an exemplary embodiment of the present disclosure;
[0026] FIG. **7** is a rear view of a display device including a first plate according to an exemplary embodiment of the present disclosure;
[0027] FIG. **8** is a cross-sectional view taken along the line II-II′ of FIG. **7** according to an exemplary embodiment of the present disclosure;
[0028] FIG. **9** is a cross-sectional view taken along the line III-III′ of FIG. **7** according to an exemplary embodiment of the present disclosure;
[0029] FIG. **10** is a cross-sectional view taken along the line IV-IV′ of FIG. **7** according to an exemplary embodiment of the present disclosure;
[0030] FIG. **11** is a plan view of a first plate as seen from a rear surface of a display panel according to another exemplary embodiment of the present disclosure;

[0031] FIG. **12**A is a plan view illustrating an example of an adhesive layer disposed below a first plate according to another exemplary embodiment of the present disclosure;

[0032] FIG. **12**B is a plan view illustrating another example of an adhesive layer disposed below a first plate according to another exemplary embodiment of the present disclosure;

[0033] FIG. **13** is a rear view of a display device including a first plate according to another exemplary embodiment of the present disclosure;

[0034] FIG. **14** is a cross-sectional view taken along the line V-V′ of FIG. **13** according to an exemplary embodiment of the present disclosure;

[0035] FIG. **15** is a cross-sectional view taken along the line VI-VI′ of FIG. **13** according to an exemplary embodiment of the present disclosure; and

[0036] FIG. **16** is a cross-sectional view taken along the line VII-VII′ of FIG. **13** according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0037] Advantages and characteristics of the present disclosure and a method of achieving the advantages and characteristics will be clear by referring to exemplary embodiments described below in detail together with the accompanying drawings. However, the present disclosure is not limited to the exemplary embodiments disclosed herein but will be implemented in various forms. The exemplary embodiments are provided by way of example only so that those skilled in the art can fully understand the disclosures of the present disclosure and the scope of the present disclosure.

[0038] The shapes, sizes, ratios, angles, numbers, and the like illustrated in the accompanying drawings for describing the exemplary embodiments of the present disclosure are merely examples, and the present disclosure is not limited thereto. Like reference numerals generally denote like elements throughout the specification. Further, in the following description of the present disclosure, a detailed explanation of known related technologies may be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure.

[0039] The terms such as “including,” “having,” and “comprising” used herein are generally intended to allow other components to be added unless the terms are used with the term “only”. Any references to singular may include plural unless expressly stated otherwise.

[0040] Components are interpreted to include an ordinary error range even if not expressly stated.

[0041] When the position relation between two parts is described using the terms such as “on”, “above”, “below”, and “next”, one or more parts may be positioned between the two parts unless the terms are used with the term “immediately” or “directly” is not used.

[0042] When an element or layer is referred to as being “on” another element or layer, it may be directly on the other element or layer, or intervening elements or layers may be present therebetween. When a component is “linked”, “coupled”, or “connected” to another component, the component may be directly linked or connected to the other component. However, unless specifically stated otherwise, it should be understood that a third component may be interposed between the components which may be indirectly linked or connected.

[0043] Although the terms “first”, “second”, and the like are used for describing various components, these components are not confined by these terms. These terms are merely used for distinguishing one component from the other components. Therefore, a first component to be mentioned below may be a second component in a technical concept of the present disclosure.

[0044] Like reference numerals generally denote like elements throughout the specification.

[0045] A size and a thickness of each component illustrated in the drawings are illustrated for convenience of explanation and are not limited to the size and the thickness of the component illustrated in embodiments of the present disclosure.

[0046] The features of various embodiments of the present disclosure can be partially or entirely coupled to or combined with each other and can be interlocked and operated in technically various ways, and respective embodiments can be carried out independently of or in association with each

other.
[0047] FIG. **1** is a block diagram of a display apparatus according to an exemplary embodiment of the present disclosure.
[0048] Referring to FIG. **1**, a display apparatus **100** according to an exemplary embodiment of the present disclosure includes an image processor **151**, a timing controller **152**, a data driver **153**, a gate driver **154**, and a display panel DP.
[0049] The image processor **151** outputs a data signal DATA supplied from the outside and a driving signal including a data enable signal DE. The image processor **151** may output a driving signal including one or more of a vertical synchronization signal, a horizontal synchronization signal, and a clock signal in addition to the data enable signal DE.
[0050] The timing controller **152** is supplied with a driving signal including a data enable signal DE and a data signal DATA from the image processor **152**. The timing controller **152** outputs a gate timing control signal GDC for controlling an operation timing of the gate driver **154** based on the driving signal. The timing controller **152** outputs a data signal DATA supplied from the image processor **151** and a data timing control signal DDC for controlling an operation timing of the data driver **153**.
[0051] The data driver **153** samples and latches the data signal DATA supplied from the timing controller **152** in response to the data timing control signal DDC supplied from the timing controller **152** to convert the data signal into a gamma reference voltage and output the converted gamma reference voltage. Further, the data driver **153** outputs a data signal through data lines DL**1** to DLn.
[0052] The gate driver **154** may output the gate signal in response to the gate timing control signal GDC supplied from the timing controller **152** and at this time, shifts a level of the gate voltage to output a gate signal. Further, the gate driver **154** outputs a gate signal through gate lines GL**1** to GLm.
[0053] The display panel DP includes a plurality of pixels P and each of the plurality of pixels P emits light in response to the data signal and the gate signal supplied from the data driver **153** and the gate driver **154**, respectively, to display images.
[0054] One pixel P is configured by a plurality of sub pixels. For example, one pixel includes three or more sub pixels which emit different color light. For example, in the display apparatus **100** according to the exemplary embodiment of the present disclosure, one pixel P includes sub pixels which emit red light, green light, and blue light. However, the number of sub pixels included in one pixel P is not limited and for example, a sub pixel which emits white light may be further included, in addition to the sub pixels which emit red light, green light, and blue light.
[0055] On the display panel DP, a plurality of gate lines GL**1** to GLm extending in a first direction and a plurality of data lines DL**1** to DLn extending in a second direction which is different from the first direction are disposed to intersect. Pixels P are defined at intersections of the plurality of gate lines and data lines on the display panel DP.
[0056] FIG. **2** is a front view of a display apparatus according to an exemplary embodiment of the present disclosure.
[0057] FIG. **3** is a plan view of a substrate of a display panel as seen from a front side according to an exemplary embodiment of the present disclosure. FIG. **4** is a cross-sectional view taken along the line I-I′ of FIG. **2** according to an exemplary embodiment of the present disclosure.
[0058] In the present specification, a direction in which the pixel P emits light on the display panel DP (that is, a direction in which an image is displayed) is referred to as a front surface of the display panel DP and an opposite direction of the direction in which the pixel P emits light is referred to as a rear surface of the display panel DP. For example, as illustrated in FIGS. **2** and **3**, a direction of the front surface of the display pane DP is defined as a Z axis and a direction of the rear surface of the display panel DP is defined as a −Z axis.
[0059] When the placement structure of each component of the display apparatus **100** is described

in the present specification, the description will be made with respect to a direction in which the image is displayed on the display panel DP. For example, components which are disposed in a front direction of the display panel DP are described to be disposed on or above the display panel DP and components which are disposed in a rear direction of the display panel DP are described to be disposed under or below the display panel DP.

[0060] As illustrated in FIGS. **2** and **4**, the display apparatus **100** includes a display panel DP including a substrate **110**, a cover layer CL disposed above the display panel DP, and an optical layer **120** and a third adhesive layer **130** disposed between the substrate **110** of the display panel DP and the cover layer CL. Further, the display apparatus **100** includes a back plate **400**, a first adhesive layer **430**, a second adhesive layer **440**, a reinforcement member **140**, a flexible film unit **150**, and a printed circuit board **500** which are disposed below the substrate **110** of the display panel DP. At this time, the back plate **400** includes a first plate **410** and a second plate **420**.

[0061] The substrate **110** is configured to support various components included in the display apparatus **100**. The substrate **110** is formed of an insulating material. Further, the substrate **110** is formed of a transparent material. Further, the substrate **110** may be a rigid substrate or a flexible substrate which is bendable, foldable, or rollable. For example, the substrate **110** may be formed of a plastic material, such as polyimide (PI). When the substrate **110** is formed of a plastic material (for example, polyimide (PI)) having flexibility, the manufacturing process of the display apparatus **100** is performed under a circumstance when a support substrate formed of glass is disposed below the substrate **110** and the support substrate may be released after completing the manufacturing process of the display apparatus **100**.

[0062] As illustrated in FIG. **3**, the substrate **110** of the display panel DP is defined by a display area DA in which a plurality of pixels P which implement an image is disposed and a non-display area NA in which a plurality of pixels P are not disposed, at an outer periphery of the display area DA. The non-display area NA of the substrate **110** is defined as a surrounding area which encloses around the display area DA, a bending area BA which extends from one side of the surrounding area and is bent toward the rear surface of the display panel DP, and a pad area PA extending from the bending area BA. Further, referring to FIG. **4**, as the substrate **110** is bent in the bending area BA, the pad area PA of the substrate **110** may be disposed to at least partially overlap the display area DA on the rear surface of the display panel DP.

[0063] In the display area DA of the substrate **110**, a plurality of sub pixels each including a light emitting diode and a thin film transistor for driving the light emitting diode and a pixel array unit including signal lines, such as a gate line GL, a data line DL, and a pixel driving power line may be disposed. Further, an encapsulation unit which covers a pixel array unit may be disposed on the substrate **110** and the encapsulation unit suppresses oxygen, moisture, and foreign materials from permeating into the light emitting diode layer. The encapsulation unit is formed with a multi-layered structure in which an organic material layer and an inorganic material layer are alternately laminated, but is not limited thereto.

[0064] In the display area DA of the substrate **110**, a display unit for displaying an image and a circuit unit for driving the display unit may be disposed.

[0065] For example, when the display apparatus **100** is an organic light emitting display apparatus, the display unit includes a light emitting diode which includes an anode, an emission layer on an anode, and a cathode on the emission layer. At this time, the display unit includes an organic emission layer as an emission layer and further includes a hole transport layer, a hole injection layer, an electron injection layer, and an electron transport layer together with the organic emission layer. However, when the display apparatus **100** is a liquid crystal display apparatus, the display unit may be configured to include a liquid crystal layer. Hereinafter, for the convenience of description, it is assumed that the display apparatus **100** is an organic light emitting display apparatus, but is not limited thereto.

[0066] The circuit unit includes various transistors, a capacitor, and wiring lines for driving the light emitting diodes and specifically, may be formed of various components, such as a driving transistor, a switching transistor, a storage capacitor, a gate line, and a data line, but is not limited thereto.

[0067] The sub pixel of the display apparatus **100** includes a switching transistor, a driving transistor, a capacitor, and a light emitting diode. The light emitting diode operates to emit light according to a driving current formed by the driving transistor. The switching transistor may perform a switching operation such that a data signal supplied through the data line DL is stored in a capacitor as a data voltage in response to a gate signal supplied through the gate line GL. The driving transistor operates to flow a constant driving current between the high potential power line and the low potential power line in response to the data voltage stored in the capacitor.

[0068] It has been described above that in the display apparatus **100** according to the exemplary embodiment of the present disclosure, one sub pixel is configured with a 2T (transistor) 1C (capacitor) structure including one switching transistor, one driving transistor, and one capacitor as an example. However, when a compensation circuit is further included, the sub pixel may be configured in various structures, such as 3T1C, 4T2C, 5T2C, 6TIC, 6T2C, 7T1C, or 7T2C. The compensation circuit is a circuit for compensating for a threshold voltage of the driving transistor and includes one or more compensating thin film transistors and compensating capacitors. At this time, configurations and structures of the compensating thin film transistor and the compensating capacitor are not limited, but may vary depending on a compensating method.

[0069] The non-display area NA of the substrate **110** is an area where no image is displayed and various wiring lines and circuits for driving the display unit disposed in the display area DA are disposed.

[0070] The pad area PA of the non-display area NA of the substrate **110** is an area where various wiring lines and circuits for receiving, with an external power, a data driving signal or transmitting and receiving a touch signal are disposed. In the pad area PA, an external module, for example, a driving IC such as a data driver IC (integrated circuit) or a gate driving IC may be located. The driving IC disposed in the pad area PA is connected to a plurality of signal lines and is connected to the plurality of data lines DL or the plurality of gate lines GL disposed in the display area DA through a plurality of signal lines. That is, the driving IC disposed in the pad area PA is electrically connected to each of the plurality of pixels P.

[0071] The bending area BA in which a part of the non-display area NA is bent in one direction is located between the display area DA and the pad area PA in the non-display area NA.

[0072] The non-display area NA is not an area where the images are displayed so that the non-display area does not need to be visible from a front surface of the display panel DP. Therefore, a partial area of the non-display area NA of the substrate **110** may be bent toward the rear surface of the display panel DP and for example, may be bent to the rear direction (that is, −Z axis) of the display panel DP so that one edge of the substrate **110** has a predetermined curvature. In this case, the pad area PA may be located so as to overlap the display area DA on the rear surface of the display panel DP. By doing this, the non-display area NA may be reduced while ensuring an area for the wiring line and the driving circuit.

[0073] The cover layer CL is disposed so as to cover the front surface of the display panel DP to protect the display panel DP from the external shock. The cover layer CL may be formed of cover glass, tempered glass, and tempered plastic, but is not limited thereto.

[0074] An edge portion of the cover layer CL may have a curvature portion or a curved portion bent in the rear direction (that is, the −Z axis) of the display panel DP. By doing this, the cover layer CL may be disposed so as to cover to a side area of the display panel DP and protect not only the front surface of the display panel DP, but also the side surface from the external shock.

[0075] The cover layer CL may be formed with a transparent material so as to overlap the display area DA of the substrate **110** of the display panel DP. For example, the cover layer CL may be formed of a cover glass of a transparent plastic material or a transparent glass material which

transmits the image, but is not limited thereto.
[0076] A front surface of the display panel DP is disposed below the cover layer CL and the display area DA and the non-display area NA of the substrate **110** of the display panel DP may be applied also to the cover layer CL in the same way. That is, the area of the cover layer CL in which the image is transmitted is a display area and an area which encloses the display area and does not allow the image to transmit may be a non-display area.
[0077] The optical area **120** is disposed on the front surface of the display panel DP and suppresses the reflection of the external light to improve visibility and a contrast ratio with respect to an image which is displayed on the display panel DP.
[0078] The third adhesive layer **130** is disposed between the cover layer CL and the optical layer **120** to bond the cover layer CL and the optical layer **120**. The third adhesive layer **130** overlaps the display area DA of the substrate **110** and uses a material which allows the image of the display panel DP to transmit. For example, the third adhesive layer **130** may be formed of an optical clear adhesive (OCA), and optical clear resin (OCR), or a pressure sensitive adhesive (PSA), but is not limited thereto.
[0079] On a lower surface of the non-display area (for example, an edge area) of the cover layer CL, a light shielding unit which blocks various circuits, wiring lines, and structures disposed in the non-display area NA of the display panel DP from being visible to the user may be disposed. The light shielding unit is formed of a material which absorbs light and for example, may be formed of a black matrix, or formed by printing with a black ink, but is not limited thereto.
[0080] The back plate **400** is disposed below the substrate **110**.
[0081] The back plate **400** may not be disposed in a portion which overlaps the bending area BA. At this time, the back plate **400** includes a first plate **410** which is disposed so as to overlap the display area DA below the substrate **110** and a second plate **420** which is disposed so as to overlap the pad area PA below the substrate **110**.
[0082] The display apparatus **100** according to the exemplary embodiment of the present disclosure includes a black layer disposed on the bottom surface of the first plate **410** to suppress the damage and the stain defect of the panel due to the exposure to the external light of the rear surface of the display panel DP. The black layer for blocking the exposure to the external light will be described in detail below with reference to FIGS. **5** to **16**.
[0083] The first plate **410** and the second plate **420** may be formed of plastic thin films having rigidity. For example, the first plate **410** and the second plate **420** may be configured by polyethylene terephthalate (PET), polyimide (PI), polyethylene naphthalate (PEN), but are not limited thereto. The first plate **410** and the second plate **420** may be formed of the same material with the same thickness, but are not limited thereto.
[0084] With respect to the shape of the substrate **110** which is not bent in the bending area BA, the first plate **410** is coupled to the display area DA of the rear surface of the substrate **110**. Therefore, the display area DA may be maintained in a flat state while supplementing the rigidity of the substrate **110**.
[0085] With respect to the shape of the substrate **110** which is not bent in the bending area BA, the second plate **420** is coupled to the pad area PA of the rear surface of the substrate **110**. Therefore, the pad area PA may be maintained in a flat state while supplementing the rigidity of the substrate **110**.
[0086] With respect to the shape of the substrate **110** which is bent in the bending area BA, the second plate **420** is disposed between the first plate **410** disposed on the display area DA of the rear surface of the substrate **110** and the pad area PA of the rear surface of the substrate **110**. At this time, the second adhesive layer **440** is disposed between the first plate **410** and the second plate **420** to bond the first plate **410** and the second plate **420** to fix the second plate **420** to the first plate **410**.
[0087] The second adhesive layer **440** may be a double-sided tape having adhesive force, but is not

limited thereto. For example, the second adhesive layer **440** is formed with a foam tape or a foam pad having adhesive force to relieve the shock.

[0088] One surface of the second adhesive layer **440** is attached onto a bottom surface of the first plate **410**. Further, in order to dispose and fix the second plate **420** below the substrate **110** of the display panel DP, the second plate **420** is connected to or attached onto the bottom surface of the pad area PA of the substrate **110** and the bending area BA of the substrate **110** is bent to attach or fix the second plate **420** to the other surface of the second adhesive layer **440**. Accordingly, the second plate **420** is disposed between the first plate **410** disposed below the display area DA of the substrate **110** and the bottom surface of the pad area PA of the substrate **110**.

[0089] When the second plate **420** is fixed to the second adhesive layer **440**, an outer portion which is a top surface of the substrate **110** in the bending area BA is exposed to the outside. Further, an inner portion which is a bottom surface of the substrate **110** is disposed so as to be opposite to a side surface of the first plate **410**, a side surface of the second adhesive layer **440**, and a side surface of the second plate **420**.

[0090] In a state in which the substrate **110** is bent in the bending area BA, a tensile force higher than that of an existing state is applied to an outer portion of the substrate **110** with a reduced radius of curvature and a compressive force higher than that of the existing state is applied to an inner portion of the substrate **110**. When the external shock is applied to the bending area BA of the substrate **110** to which the higher compressive force and tensile force are applied as described above, the bending area BA is deformed so that the substrate **110** and the signal line are cracked or damaged. In order to suppress this problem, the reinforcement member **140** may be disposed in an outer portion (that is, a top surface) of the bending area BA of the substrate **110**.

[0091] Further, in order to increase the flexibility of the display panel DP, in the bending area BA of the substrate **110**, some of other components (for example, the back plate **400**) excluding the substrate **110** and the signal line is removed so that the rigidity may be lowered. The reinforcement member **140** supplements the rigidity of the bending area BA from which other components are removed.

[0092] The reinforcement member **140** extends to the bending area BA and at least a partial area of the pad area PA to cover the front surface of the substrate **110**.

[0093] The reinforcement member **140** includes resin. For example, the reinforcement member **140** includes an ultraviolet (UV) curable acrylic resin or a thermosetting resin, but is not limited thereto, and various materials may be used. The reinforcement member **140** is formed of a cured resin that has undergone a curing process after applying the resin. For example, when the UV curable resin is used for the reinforcement member **140**, UV ray is irradiated to cure the reinforcement member and when the thermosetting resin is used for the reinforcement member **140**, heat is applied to cure the reinforcement member. For example, the reinforcement member **140** may be a micro cover layer (MCL).

[0094] Further, the reinforcement member **140** covers various signal lines disposed between an encapsulation unit which is disposed to a part of the non-display area NA from the display area DA of the substrate **110** and the pad area PA of the substrate **110**. The reinforcement member **140** protects the signal line from the external shock and suppresses the permeation of moisture.

[0095] On the pad area PA of the substrate **110**, various signal lines connected to the pixel P and the pad unit connected to each signal line are disposed. The pad unit includes a data pad which is electrically connected to the data line DL and a gate pad which is electrically connected to the gate line GL.

[0096] A controller which is electrically connected to the pad unit is disposed on the display area DA of the rear surface of the substrate **110**. The controller is mounted in the printed circuit board **500** to be disposed below the first plate **410**.

[0097] For example, the controller may be a timing controller **152**, but is not limited thereto, and may be a control device which performs various other control functions together with the timing controller **152**. The controller may be implemented by various circuits or electronic components, such as an integrated circuit (IC), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or a processor.

[0098] On the rear surface of the substrate **110**, the pad unit on the pad area PA and the controller mounted in the printed circuit board **500** on the display area DA are electrically connected through the flexible film unit **150**.

[0099] In the flexible film unit **150**, the driving IC, such as a data driver IC or a gate driver IC may be located. For example, the driving IC may be mounted in the flexible film as a chip on film (COF) type. Alternatively, the driving IC may be a flexible circuit, such as a flexible printed circuit (FPC). Further, the driving IC may be bonded directly to the pad units above the substrate **110** by the chip on glass (COG) process.

[0100] The first adhesive layer **430** is disposed between the printed circuit board **500** and the first plate **410** to bond the printed circuit board **500** and the first plate **410**. For example, the first adhesive layer **430** may be formed of a pressure sensitive adhesive (PSA), but is not limited thereto.

[0101] In the meantime, when the display panel DP is exposed to the external light from the rear surface, in the display panel, there may be a luminance deviation between an area which is consistently affected by exposed light and an area which is not affected by the exposed light. Therefore, for example, stain defects, such as shadow mura may occur. In order to suppress this problem, a black layer is disposed on a bottom surface of the first plate **410** below the display panel DP according to the exemplary embodiment of the present disclosure.

[0102] Hereinafter, a structure in which the black layer is disposed on the bottom surface of the first plate **410** will be described with reference to FIGS. **5** to **16**.

[0103] FIG. **5** is a plan view of a first plate as seen from a rear surface of a display panel according to an exemplary embodiment of the present disclosure.

[0104] Referring to FIG. **5**, a black layer **411** with a predetermined width is disposed along an edge area of the bottom surface of the first plate **410**. The black layer **411** is a light shielding layer and for example, may be a black ink layer. For example, ink including a black material which blocks light is directly printed on the bottom surface of the first plate **410** by means of a silk screen printing process to have predetermined widths (Wy, Wx) and thickness to dispose the black ink layer **411** in a predetermined position of the bottom surface of the first plate **410**. The ink material of the black ink layer **411** may have a characteristic that the ink material is not decomposed under a specific environmental condition (for example, a high temperature and high humidity environment). Hereinafter, an example that the black layer **411** according to the exemplary embodiment of the present disclosure is a black ink layer will be described.

[0105] The first plate **410** has the same or similar size to the display area DA of the substrate **110** and the surrounding area which encloses the display area DA and is disposed such that a center point of the first plate **410** is aligned with a center point of the display area DA of the substrate **110**.

[0106] A part of the edge area of the bottom surface of the first plate **410** may be an open area **410**OP which is not printed with the black ink layer **411**, but exposes the bottom surface of the first plate **410** as it is (hereinafter, referred to as a “first open area”). For example, the edge area on the bottom surface of the first plate **410** includes a first area on which the black ink layer **411** is printed and the second area (that is, a first open area **410**OP) on which the black ink layer **411** is not printed.

[0107] Further, an area excluding the edge area of the bottom surface of the first plate **410**, for example, a center area may be an open area **410**C which is not printed with the black ink layer **411**, but exposes the bottom surface of the first plate **410** as it is (hereinafter, referred to as a “second open area”).

[0108] For example, a panel alignment mark PAM disposed in one position of the rear surface of the substrate **110** of the display panel DP may be exposed to be identified, through the first open area **410**OP of the bottom surface of the first plate **410**.

[0109] The panel alignment mark PAM is located at a point exposed to the outside of the substrate, for example, is located in a corner direction of the outermost portion of the substrate **110** to be recognized by the naked eye or a camera, during the cell pre-process and post-process and a module process during the process. In FIG. **5**, an example that the panel alignment mark PAM is disposed to be close to the corner in the edge area of the rear surface of the substrate **110** is illustrated.

[0110] For example, the panel alignment mark PAM is configured by a metal layer having a specific shape and an organic film which covers the metal layer and protrudes from the rear surface of the substrate **110** with an embossing shape, but is not limited thereto.

[0111] The first plate **410** may be formed of a transparent material to transmit the panel alignment mark PAM disposed on the rear surface of the substrate **110** to be identified and for example, may be configured by a transparent polyethylene terephthalate (PET) film, but is not limited thereto.

[0112] As described above, as the black ink layer **411** is disposed so as to enclose the edge area on the bottom surface of the first plate **410**, light exposure from the rear surface of the display panel DP is blocked to suppress the panel damage and the stain defects. Further, a part of the edge area of the bottom surface of the first plate **410** is not printed with the black ink layer **411**, but is configured as the first open area **410**OP to allow the panel alignment mark PAM disposed on the rear surface of the substrate **110** of the display panel DP to transmit the first plate **410** which is a transparent material to be recognizable. By doing this, the panel alignment mark PAM required to bend the display panel DP is exposed to guide the bending of the display panel DP to suppress the misalignment of components disposed in the bending area BA of the substrate **110**. For example, when the substrate **110** is bent, a part of the flexible film unit **150** is disposed so as to overlap the first open area **410**OP of the bottom surface of the first plate **410**.

[0113] In the meantime, the first adhesive layer **430** which bonds the first plate **410** and the printed circuit board **500** is disposed in the second open area **410**C which is a center area of the bottom surface of the first plate **410**.

[0114] For example, the black ink layer **411** is disposed on the rear surface of the substrate **110** so as to overlap the surrounding area. At this time, the black ink layer **411** may be disposed along the edge of the first adhesive layer **430** to be spaced apart from the first adhesive layer **430** by a predetermined distance. That is, the widths (Wx, Wy) of the black ink layer **411** may be set depending on the size of the first adhesive layer **430**. In order to increase a light blocking efficiency from the rear surface of the display panel DP, the black ink layer **411** is set to be disposed as close as possible to the first adhesive layer **430** or overlap the outermost portion of the first adhesive layer **430** by a predetermined area.

[0115] Accordingly, the second open area **410**C of the bottom surface of the first plate **410** has the same shape as a shape of the first adhesive layer **430** and has an area almost similar to the area of the first adhesive layer **430**. By doing this, the placement position of the first adhesive layer **430** is guided below the first plate **410** and the placement position of the printed circuit board **500** which is bonded through the first adhesive layer **430** is also guided, and the misalignment while placing the first adhesive layer **430** may be suppressed. Further, the black ink layer **411** is directly printed on the bottom surface of the first plate **410** to reduce the manufacturing cost as compared with the case that a separate light shielding film, such as a black film tape is attached to the first plate **410**. Further, the misalignment between the light shielding film which is separately attached and the first adhesive layer **430** is also suppressed.

[0116] The first adhesive layer **430** disposed on the bottom surface of the first plate **410** is disposed so as to correspond to a size and a shape of the printed circuit board **500** in the second open area **410**C of the first plate **410** and may have various shapes.

[0117] FIG. **6**A is a plan view illustrating an example of an adhesive layer disposed below a first plate according to an exemplary embodiment of the present disclosure. FIG. **6**B is a plan view illustrating another example of an adhesive layer disposed below a first plate according to an exemplary embodiment of the present disclosure.

[0118] For example, as illustrated in FIG. **6**A, the first adhesive layer **430** may be disposed to have a ring shape with a predetermined width along a circumference of the second open area **410**C of the first plate **410**. As another example, as illustrated in FIG. **6**B, the first adhesive layer **430** has the same shape as a shape of the second open area **410**C of the first plate **410**. The first adhesive layer is disposed to be filled in all the areas in the second open area **410**C excluding an area in which the black ink layer **411** is disposed or disposed to be filled in all the areas spaced apart from the black ink layer **411** by a predetermined distance. Accordingly, the adhesive strength between the first adhesive layer **430** and the printed circuit board **500** may be increased. However, a size and a shape of the first adhesive layer **430** according to the exemplary embodiment of the present disclosure are not limited those illustrated in FIGS. **6**A and **6**B.

[0119] Hereinafter, a placement structure of components and a printing position of the black ink layer **411** when the substrate **110** of the display panel DP is bent in the display apparatus **100** according to the exemplary embodiment of the present disclosure will be described in more detail with reference to FIGS. **7** to **10**.

[0120] FIG. **7** is a rear view of a display device including a first plate according to an exemplary embodiment of the present disclosure.

[0121] FIG. **8** is a cross-sectional view taken along the line II-II′ of FIG. **7** according to an exemplary embodiment of the present disclosure, FIG. **9** is a cross-sectional view taken along the line III-III′ of FIG. **7** according to an exemplary embodiment of the present disclosure, and FIG. **10** is a cross-sectional view taken along the line IV-IV′ of FIG. **7** according to an exemplary embodiment of the present disclosure.

[0122] Referring to FIGS. **7** and **8** together, in the plan view, in an area which overlaps the first open area **410**OP along one direction (that is, a Y-direction) on the bottom surface of the first plate **410**, the black ink layer **411** is printed only in one edge area of the bottom surface of the first plate **410**. The remaining area is an open area through which the bottom surface of the first plate **410** is exposed.

[0123] However, in the second open area **410**C in which the black ink layer **411** is not printed in the first plate **410**, the first adhesive layer **430** and the printed circuit board **500** are disposed so that the external light from the rear surface of the display panel DP is not incident or minimized in the corresponding area.

[0124] Further, in the first open area **410**OP in which the black ink layer **411** is not printed in the first plate **410**, the second adhesive layer **440**, the second plate **420**, the pad area PA of the substrate **110**, and a part of the flexible film unit **150** which is connected between the pad area PA of the substrate **110** and the printed circuit board **500** are disposed to overlap. Accordingly, the external light from the rear surface of the display panel DP is not incident or minimized. At this time, the size of the first open area **410**OP may be minimized in a state in which the area through which the panel alignment mark PAM is exposed is ensured. For example, in an edge area including the first open area **410**OP of the first plate **410**, the black ink layer **411** is disposed as close as possible to the pad area PA of the substrate **110** and the flexible film unit **150** or is disposed to overlap the outermost portions of the pad area PA of the substrate **110** and the flexible film unit **150** by a predetermined area.

[0125] Referring to FIGS. **7** and **9** together, in the plan view, in an area which overlaps an area between the first open area **410**OP and the outermost portion of the first plate **410** along one direction (that is, a Y-direction) on the bottom surface of the first plate **410**, the black ink layer **411** is printed only in one side of one edge area of the bottom surface of the first plate **410** and the other side of the edge area opposite thereto. The remaining area is an open area through which the bottom surface of the first plate **410** is exposed.

[0126] However, in the second open area **410**C in which the black ink layer **411** is not printed in the first plate **410**, the first adhesive layer **430** and the printed circuit board **500** are disposed so that the external light from the rear surface of the display panel DP is not incident or minimized in the corresponding area.

[0127] Referring to FIGS. **7** and **10** together, in the plan view, in an area overlapping an area of the outermost portion of the first plate **410** along one direction (that is, the Y direction) on the bottom surface of the first plate **410**, the black ink layer **411** is printed in the entire edge area of the bottom surface of the first plate **410**.

[0128] As described above, the black ink is directly printed in the edge area of the bottom surface of the first plate **410** disposed on the rear surface of the display panel DP to dispose the black ink layer **411**. By doing this, the exposure to the external light from the rear surface of the display panel DP is blocked to suppress the panel damage and the stain defects. Further, the first plate **410** is formed of a transparent material to transmit the panel alignment mark PAM disposed on the bottom surface of the display panel DP to be identified and an open area in which the black ink layer **411** is not formed in an area which overlaps the panel alignment mark PAM is included in the edge area of the first plate **410**. Therefore, the panel alignment mark PAM required to bend the display panel DP is exposed to guide the bending of the display panel DP to suppress the misalignment of the components disposed in the bending area BA of the substrate **110**. Further, the black ink layer **411** is disposed along a shape of the first adhesive layer **430** in the edge area of the bottom surface of the first plate **410** to guide the placement position of the first adhesive layer **430** below the first plate **410**, thereby suppressing the misalignment when the first adhesive layer **430** is disposed.

[0129] The shape of the black ink layer **411** which is printed on the bottom surface of the first plate **410** is not limited to that described with reference to FIGS. **5** to **10**.

[0130] Hereinafter, a structure in which the black ink layer is disposed in the first plate of the display apparatus according to another exemplary embodiment of the present disclosure will be described in detail. However, the display apparatus according to another exemplary embodiment of the present disclosure includes components and features thereof which are substantially the same as or similar to those of the display device according to the exemplary embodiment of the present disclosure except for the structure in which the black ink layer is disposed on the first plate. Therefore, for the convenience of description, description of duplicate components and features thereof will be omitted. Further, among components of the display apparatus according to another exemplary embodiment of the present disclosure illustrated in FIGS. **11** to **16**, components denoted by the same reference numeral as the components of the display apparatus according to the exemplary embodiment of the present disclosure which have been described above in FIGS. **5** to **10** may refer to the substantially same components.

[0131] FIG. **11** is a plan view of a first plate as seen from a rear surface of a display panel according to another exemplary embodiment of the present disclosure.

[0132] Referring to FIG. **11**, a black ink layer **411**′ is disposed in the entire bottom surface of the first plate **410**′ except for the first open area **410**OP′ included in one side of the edge area. For example, as compared with the first plate **410** according to one exemplary embodiment of the present disclosure which has been described above with reference to FIG. **5**, the black ink layer **411**′ is disposed not only in the edge area of the bottom surface of the first plate **410**′ except for the first open area **410**OP′, but also in the second open area **410**C which is a center area. Accordingly, the first adhesive layer **430** is disposed on the bottom surface of the black ink layer **411**′ on the bottom surface of the first plate **410**′. At this time, in the plan view, there is no interval between the black ink layer **411**′ and the first adhesive layer **430**. As described above, as the black ink layer **411**′ is disposed on the entire bottom surface of the first plate **410**′ except for the first open area **410**OP′, the convenience during the printing process of the black ink layer **411**′ is increased and the external light blocking efficiency from the rear surface of the display panel DP is further increased.

[0133] Further, as illustrated in FIG. **11**, in the first open area **410**OP′ of the first plate **410**′, the black ink layer **411**′ is not printed, but the bottom surface of the first plate **410**′ is exposed as it is.

Therefore, the panel alignment mark PAM disposed in one position of the rear surface of the substrate **110** of the display panel DP may be identified through the first open area **410**OP′.

[0134] The first adhesive layer **430** disposed on the bottom surface of the first plate **410′** is disposed so as to correspond to a size and a shape of the printed circuit board **500** in a center area having a predetermined area based on the center point of the first plate **410′** and may have various shapes.

[0135] FIG. **12**A is a plan view illustrating an example of an adhesive layer disposed below a first plate according to another exemplary embodiment of the present disclosure. FIG. **12**B is a plan view illustrating another example of an adhesive layer disposed below a first plate according to another exemplary embodiment of the present disclosure.

[0136] For example, as illustrated in FIG. **12**A, the first adhesive layer **430** may be disposed to have a ring shape with a predetermined width in a position spaced apart from an edge of the first plate **410** by a predetermined interval. At this time, a partial area close to the center point on the bottom surface of the first plate **410′** is exposed from the first adhesive layer **430**. As another example, as illustrated in FIG. **12**B, the first adhesive layer **430** may be disposed to be filled in all the area in a position spaced apart along an edge of the first plate **410** by a predetermined interval. However, a size and a shape of the first adhesive layer **430** according to another exemplary embodiment of the present disclosure are not limited to those illustrated in FIGS. **12**A and **12**B.

[0137] Hereinafter, a placement structure of components and a printing position of the black ink layer **411′** when the substrate **110** of the display panel DP is bent in the display apparatus according to another exemplary embodiment of the present disclosure will be described in more detail with reference to FIGS. **13** to **16**.

[0138] FIG. **13** is a rear view of a display device including a first plate according to another exemplary embodiment of the present disclosure.

[0139] FIG. **14** is a cross-sectional view taken along the line V-V′ of FIG. **13** according to an exemplary embodiment of the present disclosure, FIG. **15** is a cross-sectional view taken along the line VI-VI′ of FIG. **13** according to an exemplary embodiment of the present disclosure, and FIG. **16** is a cross-sectional view taken along the line VII-VII′ of FIG. **13** according to an exemplary embodiment of the present disclosure.

[0140] Referring to FIGS. **13** and **14** together, in an area overlapping the first open area **410**OP′ along one direction (that is, the Y direction) on the bottom surface of the first plate **410′**, the black ink layer **410** is printed not only in the edge area of the bottom surface of the first plate **410′**, but also in the entire surface excluding the first open area **410**OP′.

[0141] In the first open area **410**OP′ in which the black ink layer **411′** is not printed in the first plate **410′**, the second adhesive layer **440**, the second plate **420**, the pad area PA of the substrate **110**, and a part of the flexible film unit **150** which is connected between the pad area PA of the substrate **110** and the printed circuit board **500** are disposed to overlap. At this time, at least one of the second adhesive layer **440**, the second plate **420**, the pad area PA of the substrate **110**, and the flexible film unit **150** which is connected between the pad area PA of the substrate **110** and the printed circuit board **500** may at least partially overlap the black ink layer **411′** in a portion excluding a portion overlapping the first open area **410**OP′ in the plan view.

[0142] Referring to FIGS. **13** and **15** together, in the plan view, in the entire area overlapping an area between the first open area **410**OP′ and an outermost portion of the first plate **410′** along one direction (that is, the Y direction) on the bottom surface of the first plate **410′**, the black ink layer **411′** is printed.

[0143] Referring to FIGS. **13** and **16** together, in the plan view, also in an entire area overlapping an area of the outermost portion of the first plate **410′** along one direction (that is, the Y direction) on the bottom surface of the first plate **410′**, the black ink layer **411′** is printed.

[0144] As described above, the black ink is directly printed in the entire surface excluding a partial area included in one side of the edge area of the bottom surface of the first plate **410′** disposed on the rear surface of the display panel DP to dispose the black ink layer **411′**. By doing this, the exposure to the external light from the rear surface of the display panel DP is blocked to suppress the panel damage and the stain defects. Further, the first plate **410′** is formed of a transparent material to transmit the panel alignment mark PAM disposed on the rear surface of the display panel DP to be identified and an open area in which the black ink layer **411′** is not formed in an area which overlaps the panel alignment mark PAM is included in one side of the edge area of the first plate **410′**. Therefore, the panel alignment mark PAM required to bend the display panel DP is exposed to guide the bending of the display panel DP, thereby suppressing the misalignment of the components disposed in the bending area BA of the substrate **110**.

[0145] The display apparatus according to the exemplary embodiment of the present disclosure may be applicable to a mobile device, a video phone, a smart watch, a watch phone, a wearable apparatus, a foldable apparatus, a rollable apparatus, a bendable apparatus, a flexible apparatus, a curved apparatus, a sliding apparatus, a variable apparatus, a personal digital assistant, an electronic book, a portable multimedia player (PMP), a personal digital assistant (PDA), an MP3 player, a mobile medical apparatus, a desktop PC, a laptop PC, a netbook computer, a workstation, a navigation, a navigation for a vehicle, a display apparatus for a vehicle, an apparatus for a vehicle, a theatrical device, a theatrical display, a television, a wallpaper device, a signage device, a game device, a notebook, a monitor, a camera, a camcorder, and a consumer electronics device. Further, the display apparatus according to the exemplary embodiment of the present disclosure is also applied to an organic light emitting illumination device or an inorganic light emitting illumination device.

[0146] The exemplary embodiments of the present disclosure can also be described as follows:

[0147] According to an aspect of the present disclosure, there is provided a display apparatus. The display apparatus includes a display panel including a substrate in which a display area in which a plurality of pixels is disposed and a non-display area in an outer periphery of the display area are defined. The display apparatus further includes a first plate disposed below the display panel. The display apparatus further includes a black layer disposed on a bottom surface of the first plate. The display apparatus further includes a printed circuit board disposed below the first plate. The display apparatus further includes a first adhesive layer which bonds the first plate and the printed circuit board. A bottom surface of the first plate includes a center area and an edge area which encloses the center area and includes a first area and a second area and the black layer is disposed in the first area among the first area and the second area.

[0148] The black layer may be a black ink layer in which an ink including a black material is printed on a bottom surface of the first plate.

[0149] The black layer may be disposed in the first area of the edge area among the center area and the edge area.

[0150] The first adhesive layer may be disposed in at least a part of the center area.

[0151] The first adhesive layer may be disposed in the center area to have a ring shape.

[0152] In the plan view, the black layer may be spaced apart from an edge of the first adhesive layer by a predetermined interval.

[0153] The black layer may be further disposed in the center area.

[0154] The first adhesive layer may be disposed on a bottom surface of the black layer disposed in the center area.

[0155] The first adhesive layer may be disposed in the center area to have a ring shape.

[0156] The display panel may include a panel alignment mark disposed on the bottom surface of the substrate and the panel alignment mark may be disposed so as to overlap the second area.

[0157] The non-display area of the substrate may include a surrounding area which encloses the display area, a bending area which extends from one side of the surrounding area and is bent, and a pad area which extends from the bending area and is disposed below the display area as the substrate is bent in the bending area. The display apparatus may further include a second plate

which is disposed in the pad area below the display panel; and a second adhesive layer which bonds the second plate and the first plate.
[0158] The display apparatus may further include: a flexible film unit which electrically connects a controller disposed in the printed circuit board and the pad unit disposed on the pad area of the substrate.
[0159] The flexible film unit may be disposed so as to overlap the second area.
[0160] Although the exemplary embodiments of the present disclosure have been described in detail with reference to the accompanying drawings, the present disclosure is not limited thereto and may be embodied in many different forms without departing from the technical concept of the present disclosure. Therefore, the exemplary embodiments of the present disclosure are provided for illustrative purposes only but not intended to limit the technical concept of the present disclosure. The scope of the technical concept of the present disclosure is not limited thereto. Therefore, it should be understood that the above-described exemplary embodiments are illustrative in all aspects and do not limit the present disclosure. The protective scope of the present disclosure should be construed based on the following claims, and all the technical concepts in the equivalent scope thereof should be construed as falling within the scope of the present disclosure.

Claims

1. A display apparatus, comprising: a display panel including a substrate, the substrate including a display area in which a plurality of pixels are disposed and a non-display area in an outer periphery of the display area; a first plate below the display panel; a black layer on a bottom surface of the first plate; a printed circuit board below the first plate; and a first adhesive layer bonding the first plate and the printed circuit board, wherein a bottom surface of the first plate includes a center area and an edge area which encloses the center area and includes a first area and a second area, wherein the black layer is in the first area among the first area and the second area.
2. The display apparatus according to claim 1, wherein the black layer is a black ink layer in which an ink including a black material is printed on the bottom surface of the first plate.
3. The display apparatus according to claim 1, wherein the black layer is in the first area of the edge area among the center area and the edge area.
4. The display apparatus according to claim 3, wherein the first adhesive layer is in at least a part of the center area.
5. The display apparatus according to claim 4, wherein the first adhesive layer is in the center area and has a ring shape.
6. The display apparatus according to claim 4, wherein in a plan view, the black layer is spaced apart from an edge of the first adhesive layer by a predetermined interval.
7. The display apparatus according to claim 4, wherein in a plan view of the display apparatus, the black layer overlaps an outermost portion of the first adhesive layer by a predetermined area.
8. The display apparatus according to claim 1, wherein the black layer is disposed in the center area and the first area of the edge area among the center area and the edge area.
9. The display apparatus according to claim 8, wherein the first adhesive layer is on a bottom surface of the black layer disposed in the center area.
10. The display apparatus according to claim 9, wherein the first adhesive layer is in the center area and has a ring shape.
11. The display apparatus according to claim 1, wherein the display panel includes a panel alignment mark on a bottom surface of the substrate and the panel alignment mark overlaps the second area of the edge area.
12. The display apparatus according to claim 11, wherein the first plate includes a transparent material.
13. The display apparatus according to claim 1, wherein the non-display area of the substrate

includes a surrounding area enclosing the display area, a bending area extending from one side of the surrounding area and bending, and a pad area extending from the bending area and is below the display area as the substrate is bent in the bending area, the display apparatus further comprising: a second plate in the pad area below the display panel; and a second adhesive layer bonding the second plate and the first plate.

14. The display apparatus according to claim 13, further comprising: a flexible film unit electrically connecting a controller disposed in the printed circuit board and a pad unit disposed on the pad area of the substrate.

15. The display apparatus according to claim 14, wherein the flexible film unit overlaps the second area.

16. The display apparatus according to claim 13, further comprising: a reinforcement member in an outer portion of the bending area.